University of Pittsburgh | School of Public Health

# More statistical learning tools
# - Tree-based methods
# - Bagging, Boosting, and Random forests

## BIOST 2155

**Class 10**
**November 6th, 2026**

**Statistical modeling**: The **two cultures** (with comments and a rejoinder by the author)

L Breiman - **Statistical** science, 2001 - projecteuclid.org

… There are **two cultures** in the use of **statistical modeling** to reach conclusions from data. One … **model**. The other uses algorithmic **models** and treats the data mechanism as unknown. The …

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[PDF] projecteuclid.org

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# Assigned reading (not compulsory)



## Leo Breiman 1928-2005

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Professor of Statistics, UC Berkeley
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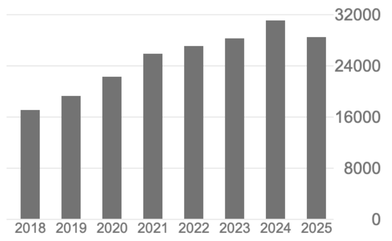
Data Analysis  Statistics  Machine Learning

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| TITLE | CITED BY | YEAR |
|---|---|---|
| Random forests<br>L Breiman<br>Machine learning 45 (1), 5-32 | 167626 | 2001 |
| Classification and Regression Trees<br>L Breiman, JH Friedman, RA Olshen, CJ Stone<br>CRC Press, New York | ~~70583~~ * | 1999 |
| Classification and regression trees<br>L Breiman<br>Chapman & Hall/CRC | 70583 * | 1984 |
| Bagging predictors<br>L Breiman<br>Machine learning 24 (2), 123-140 | 41191 | 1996 |

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| Citations | 316213 | 163048 |
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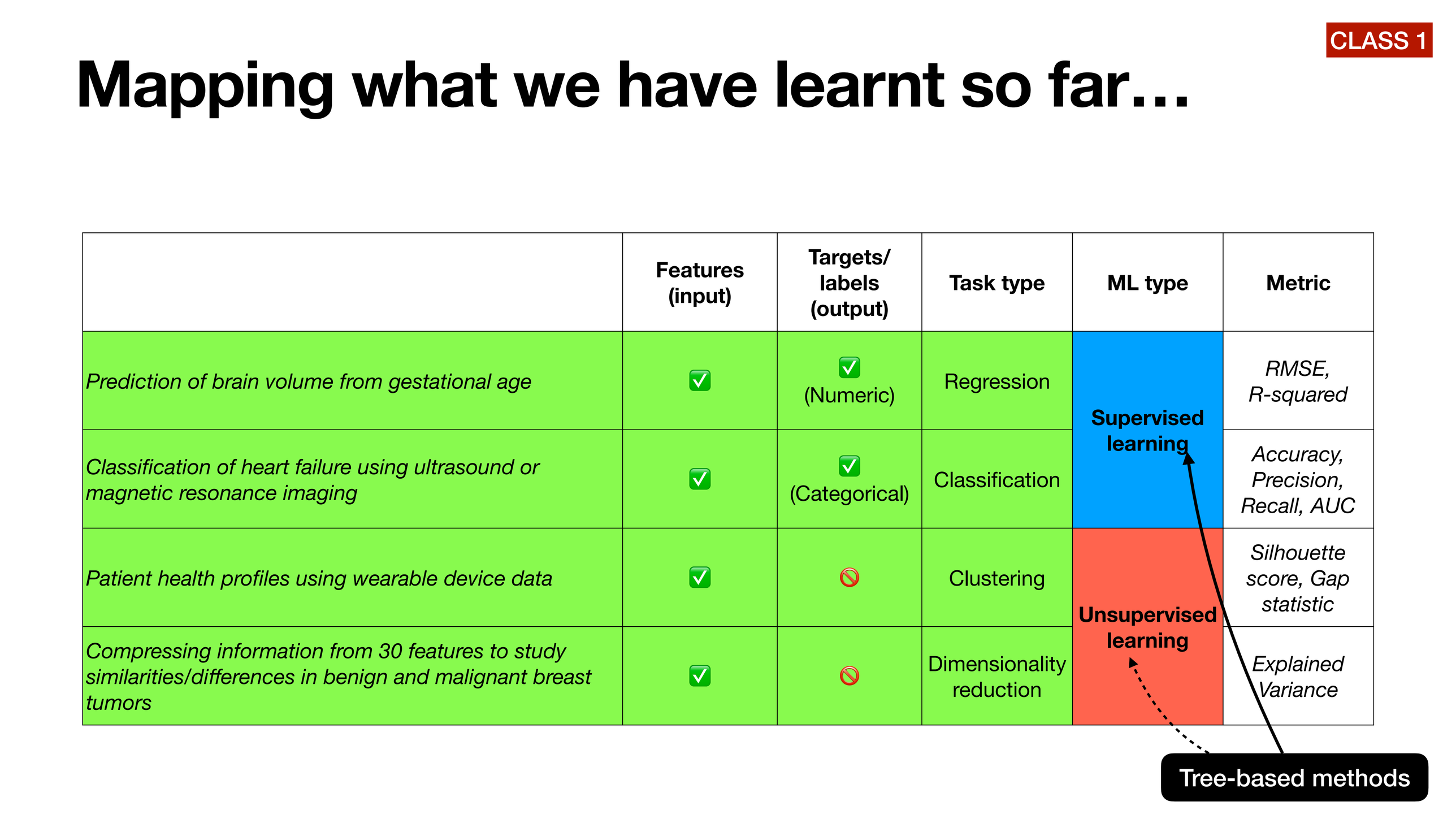



In 2003

CLASS 1

# Mapping what we have learnt so far…

| | Features (input) | Targets/ labels (output) | Task type | ML type | Metric |
|---|---|---|---|---|---|
| *Prediction of brain volume from gestational age* | ✅ | ✅ (Numeric) | Regression | **Supervised learning** | *RMSE, R-squared* |
| *Classification of heart failure using ultrasound or magnetic resonance imaging* | ✅ | ✅ (Categorical) | Classification | | *Accuracy, Precision, Recall, AUC* |
| *Patient health profiles using wearable device data* | ✅ | 🚫 | Clustering | **Unsupervised learning** | *Silhouette score, Gap statistic* |
| *Compressing information from 30 features to study similarities/differences in benign and malignant breast tumors* | ✅ | 🚫 | Dimensionality reduction | | *Explained Variance* |



# Tree-based methods

## Overview

- Classical tree-based methods can be applied for supervised learning: regression and classification tasks.
- Newer tree based methods have been developed for unsupervised learning as well
    - Isolation forests[1]: unsupervised anomaly/outlier detection
    - Unsupervised random forests[2]: applying random forests to derive distances between data points, then use for clustering

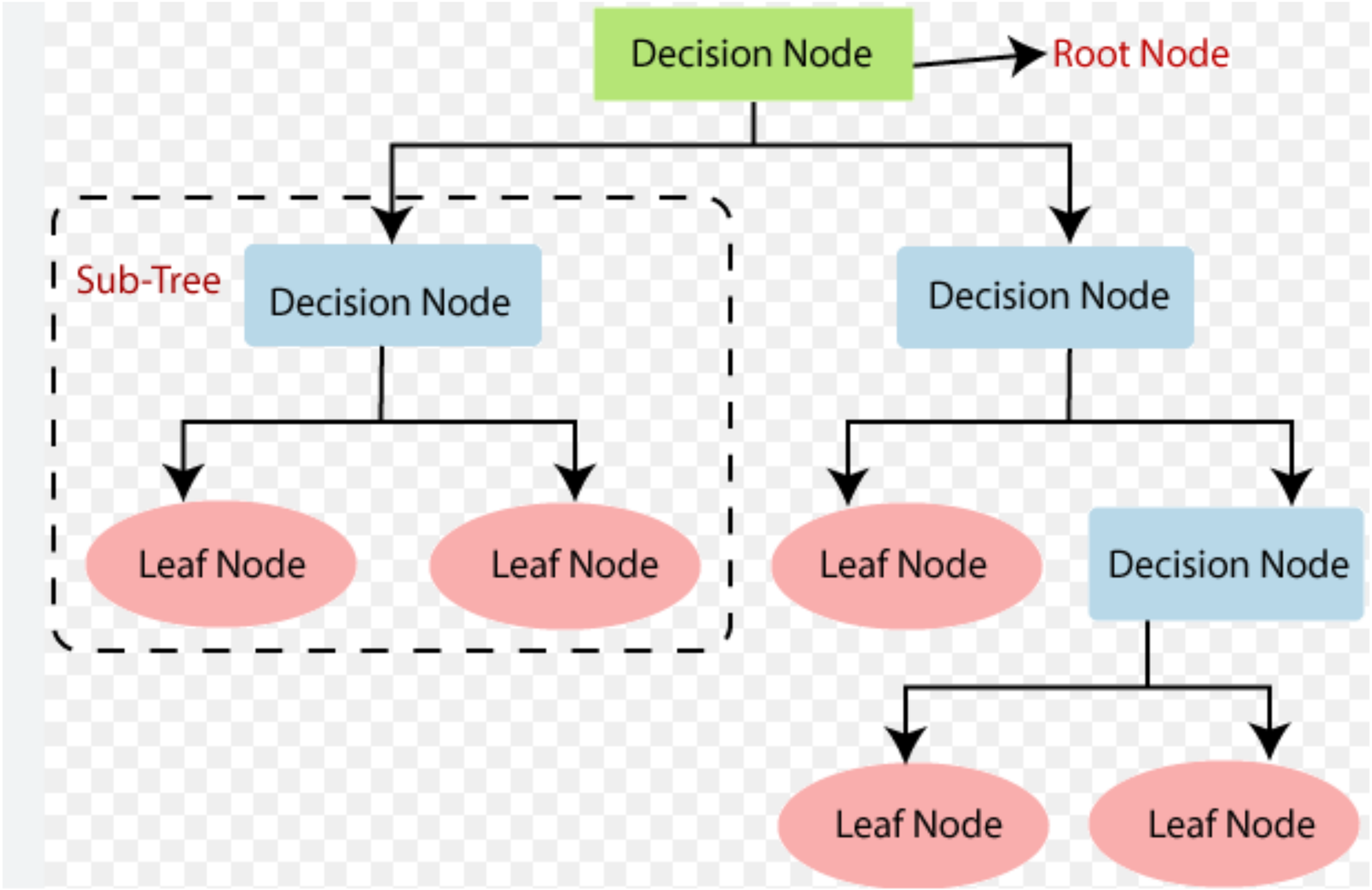
**Involve stratifying/segmenting the feature space into some simple regions.**


1. F. T. Liu, K. M. Ting and Z. -H. Zhou, "Isolation Forest," 2008 Eighth IEEE International Conference on Data Mining, Pisa, Italy, 2008, pp. 413-422, doi: 10.1109/ICDM.2008.17

2. Shi, Tao, and Steve Horvath. "Unsupervised learning with random forest predictors." Journal of Computational and Graphical Statistics 15.1 (2006): 118-138.

# Tree-based methods

## Terminology

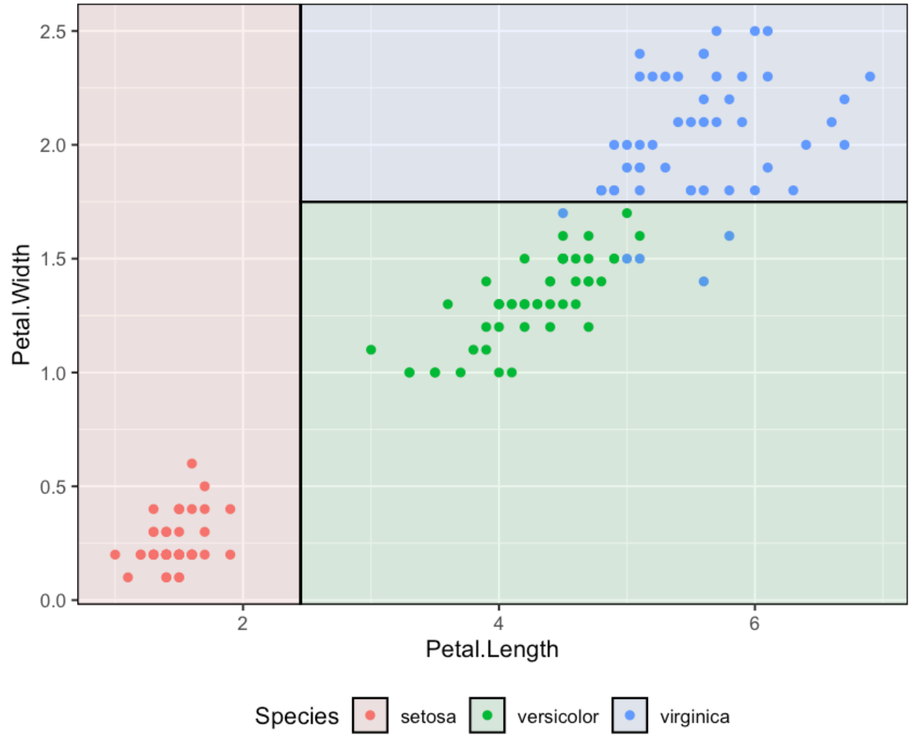


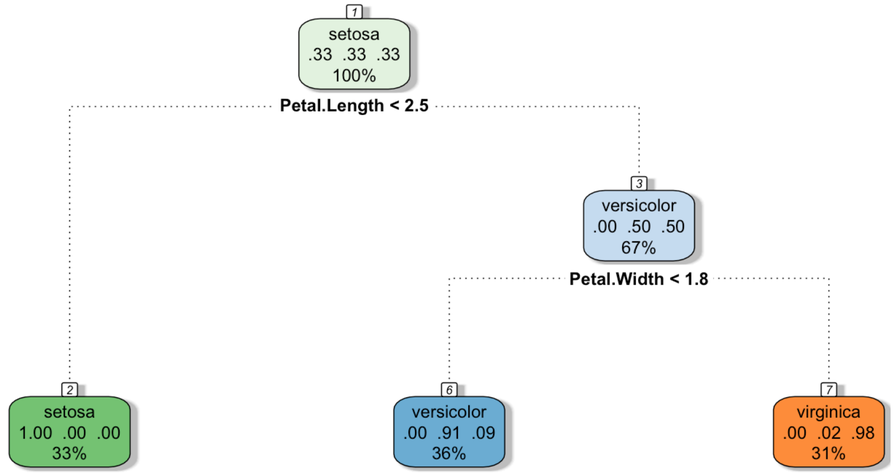
# Tree-based methods

**Strengths, weaknesses, and scope for improvement…**

- **Tree-based methods are simple and useful for interpretation.**
- However they typically are not competitive with the best supervised learning approaches in terms of prediction accuracy.

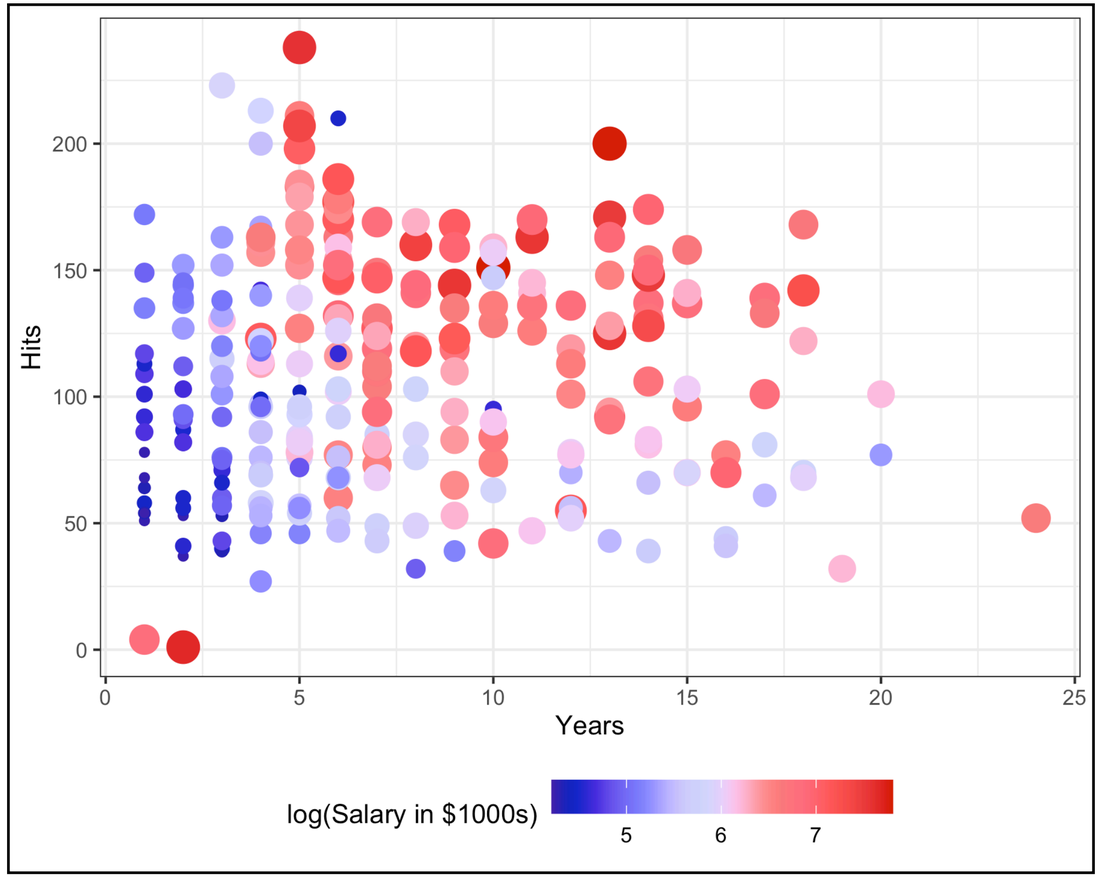
- **Hence we also discuss bagging, random forests, and boosting. These methods grow multiple trees which are then combined to yield a single consensus prediction.**
- Combining a large number of trees can often result in dramatic improvements in prediction accuracy, at the expense of loss of interpretation.

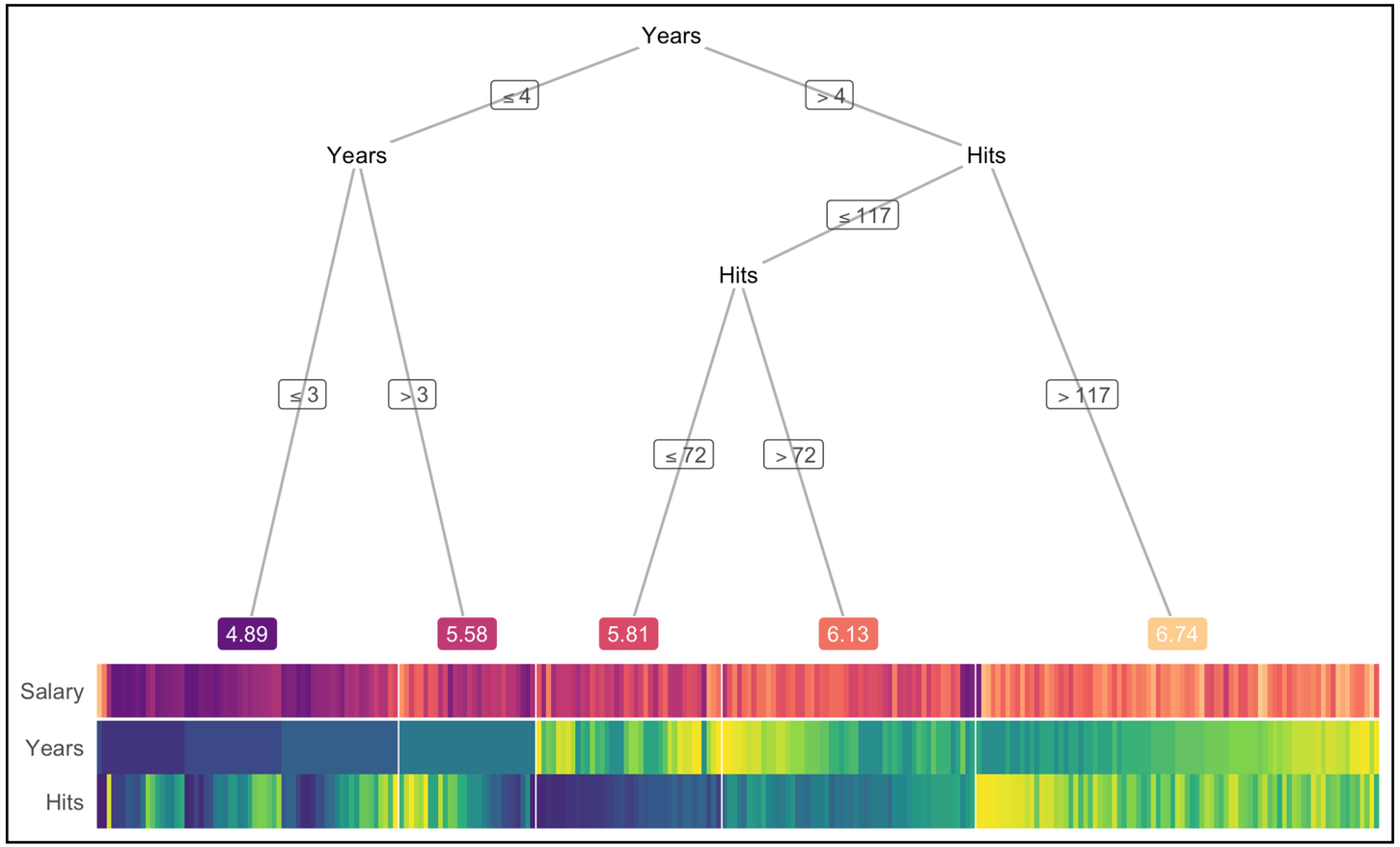


# Regression trees

## Predicting Baseball Players' Salaries Using Regression Trees



# Prediction via Stratification

## Details of the regression tree-building process

- We divide the predictor space — that is, the set of possible values for $X_1, X_2, \ldots, X_p$ — into $J$ distinct and non-overlapping regions $R_1, R_2, \ldots, R_J$
- For observations in the region $R_j$, we make the same prediction: the mean of the response for all training data in $R_j$:

$$\hat{y}_{R_j} = \text{ave}_{i \in R_j} y_i$$

- **How to define regions $R_1, R_2, \ldots, R_J$?**
- **How to quantify model performance?**

# Prediction via Stratification

## More details of the regression tree-building process

- The regions $R_1, R_2, \ldots, R_J$ could have **any** shape.
- Divide the predictor space into $p$**-dimensional hypercubes** for simplicity and for ease of interpretation.
    - For $p = 1$: intervals; for $p = 2$: rectangles; for $p = 3$ cubes….
- In specialized circumstances, can think of concentric circles, spheres, hyperspheres: epidemiology, climate sciences, etc.

Optimization problem: find regions $R_1, R_2, \ldots, R_J$ that minimize:

$$RSS = \sum_{j=1}^{J} \sum_{i \in R_j} \left( y_i - \hat{y}_{R_j} \right)^2$$

# Prediction via Stratification

## EVEN MORE details of the regression tree-building process

- Computationally infeasible to consider every possible partition of the feature space into $J$ boxes.
- Take a **top-down**, **greedy** approach: **recursive binary splitting**.
    - **Top-down**: begins at the top of the tree and then successively splits the predictor space; each split is indicated via two new branches further down on the tree.
    - **Greedy:** at each step, the best split is made at that particular step, rather than looking ahead and picking a split that will lead to the best tree in some future step.

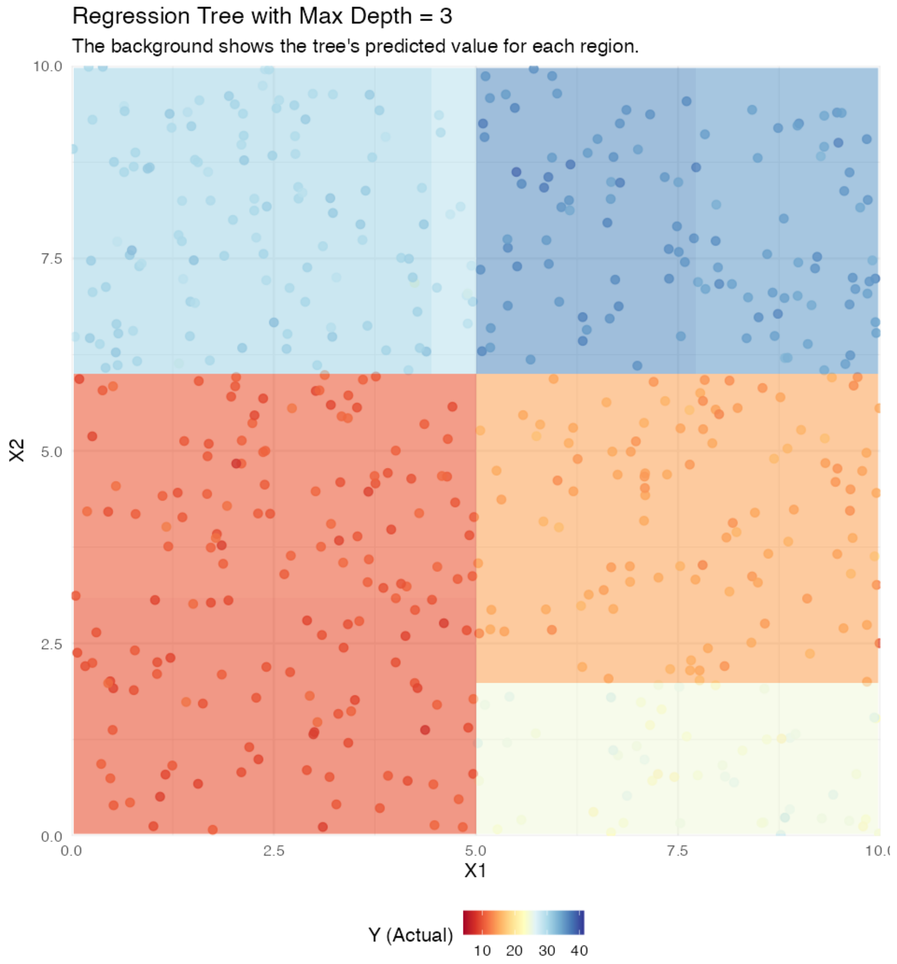
Optimization problem: find regions $R_1, R_2, \ldots, R_J$ that minimize:

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# Prediction via Stratification

## EVEN MORE (x2) details of the regression tree-building process

- Step 1: select $X_j$ and cut-point $s$ such that splitting the predictor space into the regions $R_1(j,s) = \left\{X \,|\, X_j < s\right\}$ and $R_2(j,s) = \left\{X \,|\, X_j \geq s\right\}$ leads to the greatest possible reduction in $RSS$.
- Step 2: look for the best predictor and best cut-point in order to split the data further so as to minimize the RSS within each of $R_1(j,s)$ and $R_2(j,s)$.
    - Instead of splitting the entire space, split **only one** of $R_1(j,s)$ and $R_2(j,s)$: now have three regions.
- Step 3: we look to split **only one** of these three regions further.
- Repeat until a **stopping criterion** is reached: until no region contains more than five observations (‘minimum points per node’) or until a tree of some pre-specified ‘depth’ has been constructed.



# Pruning a tree

## Trees struggle a lot with overfitting

- Produces good predictions on the training set, but is likely to overfit the data, leading to poor test set performance - because the resulting tree $T_0$ might be too complex.
- A smaller tree $T_1$ with fewer splits might lead to lower variance and better interpretation at the cost of a little bias.
- Mitigation strategy: grow a very large tree $T_0$, and then **prune** it back in order to obtain a subtree. Cost complexity pruning — also known as weakest link pruning — is used to do this.



Large tree with low bias but (potentially) high variance because of overfitting



Pruned tree - more bias but lower variance, better overall MSE

# Pruning a tree

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1. Start with $T_0$ and specify tuning parameter $\alpha$
2. Find $T_1 \subset T_0$ that minimizes

$$\sum_{m=1}^{|T_1|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T_1|$$

$|T_1|$: number of split-up regions in $T_1$

$R_m$: split-up regions

$\hat{y}_{R_m}$: mean of training observations in $R_m$

**Choose $\alpha$ using k-fold CV.**

**Larger $\alpha$ will give you smaller trees!**

# Prediction via Stratification

## Details of the *classification* tree-building process

- Very similar to a regression tree, except that it is used to predict a qualitative response rather than a quantitative one.
- Predict that each observation belongs to the **most commonly occurring class** of training observations in the region to which it belongs.
- Need alternatives to $RSS$ within $R_m$: we define $\hat{p}_{mk}$ = proportion of training observations in $R_m$ that are from the $k$-th class

Misclassification rate: $E = 1 - \max_k \hat{p}_{mk}$

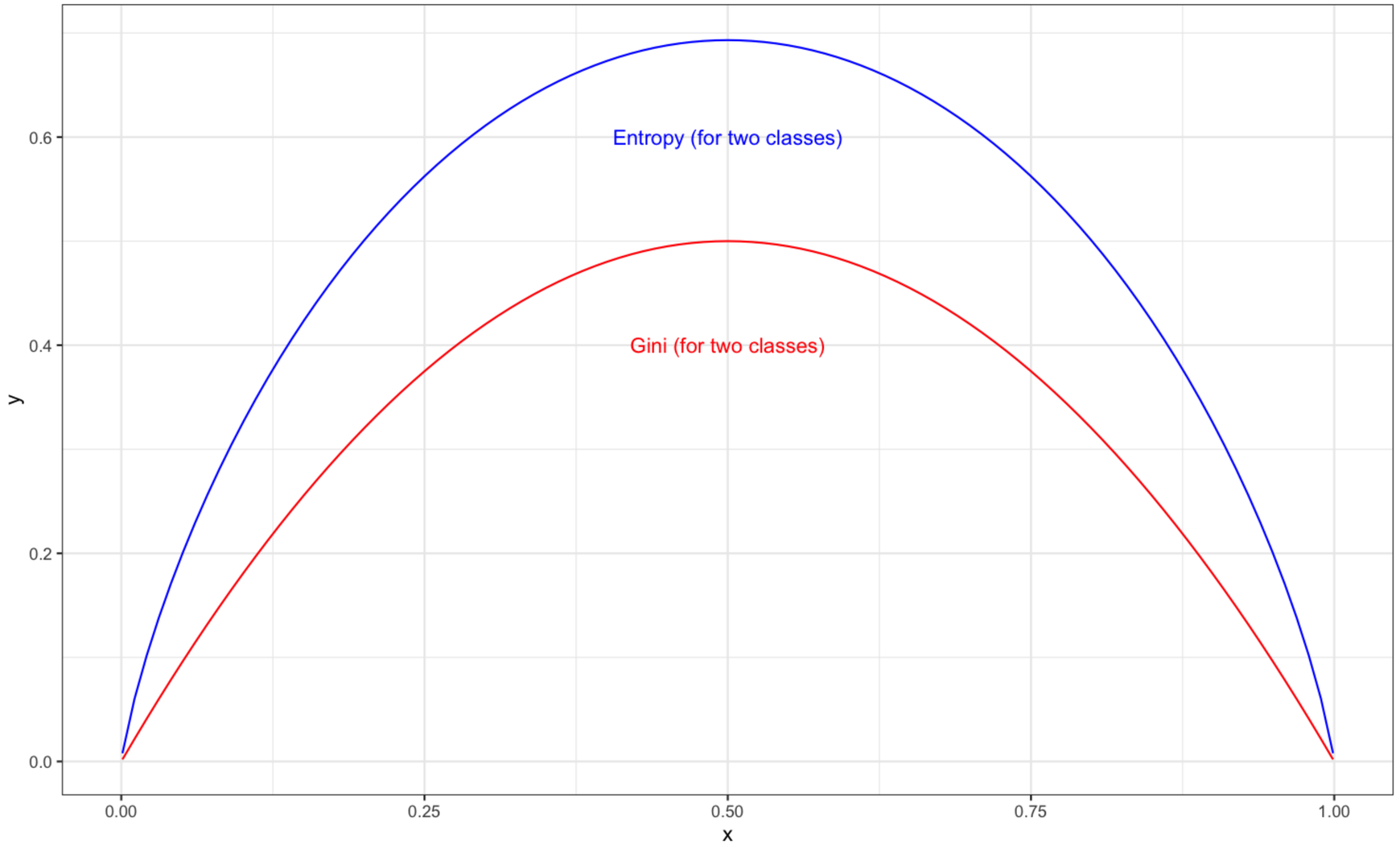
Gini index: $G = \sum_{k=1}^{K} \hat{p}_{mk}\left(1 - \hat{p}_{mk}\right)$

Entropy: $D = -\sum_{k=1}^{K} \hat{p}_{mk} \log(\hat{p}_{mk})$

Not sufficiently sensitive

Measures of impurity in node

We will still make use of recursive binary pruning, using $G$ or $D$




Misclassification rate: $E = 1 - \max_k \hat{p}_{mk}$

Gini index: $G = \sum_{k=1}^{K} \hat{p}_{mk}\left(1 - \hat{p}_{mk}\right)$

Entropy: $D = -\sum_{k=1}^{K} \hat{p}_{mk} \log(\hat{p}_{mk})$

**Not sufficiently sensitive**

**Measures of impurity in node**

# Advantages and Disadvantages of Trees

- Trees are very easy to explain to people: easier to explain than linear regression!
- Decision trees more closely mirror human decision-making than do the regression and classification approaches.
- Trees can be displayed graphically, and are easily interpreted even by a non-expert (especially if they are small).
- Trees can easily handle qualitative predictors without the need to create dummy variables.

- Trees generally do not have the same level of predictive accuracy as most other regression and classification approaches.
- A small change in the data can cause a large change in the final estimated tree

**Opportunity: By aggregating many decision trees, the predictive performance of trees can be substantially improved….**

# Improvement 1: BAGging

## BAG = Bootstrap Aggregation

- General-purpose procedure for reducing the variance of ANY statistical learning method; particularly useful in the context of decision trees.
- Given a set of n independent observations $Z_1, Z_2, \ldots, Z_n$ each with variance $\sigma^2$, the variance of the mean $\bar{Z}$ is $\sigma^2/n$ - which means averaging a set of observations reduces variance…but we don't have multiple training sets!

- Bootstrap (sample rows with replacement) $B$ different training sets.
- Train models on each bootstrapped set, call them $\hat{f}_b(\boldsymbol{x}),\ 1 \leq b \leq B$

**Aggregate all models** $\hat{f}_{\mathbf{bag}}(\boldsymbol{x}) = B^{-1} \sum_{b=1}^{B} \hat{f}_b(\boldsymbol{x})$

Classification trees: get $\boldsymbol{x}$, record $\hat{f}_b(\boldsymbol{x})$, take a majority vote.

# Improvement 1: BAGging

## Out-of-Bag error estimation

- Recall that the **key to bagging** is that trees are **repeatedly fit to bootstrapped subsets of the observations**.
  - One can show that on average, each bagged tree makes use of around two-thirds of the observations….recall $0.632$ bootstrap!
- **The remaining one-third of the observations not used to fit a given bagged tree are referred to as the out-of-bag (OOB) observations. No extra computation cost!**
- We can predict the response for the $i$th observation using each of the trees in which that observation was OOB.
  - **This will yield around $B/3$ predictions for the $i$th observation, which we average.**
- This estimate is essentially the LOO cross-validation error for bagging, if $B$ is large.

# Improvement 2: Boosting

## Boost model performance by learning from prior errors

- Boosting is similar to bagging - we bootstrap $B$ training samples of size $n$
- Bagging involves creating multiple copies of the original training data set using the bootstrap, fitting a separate decision tree to each copy, and then combining all of the trees in order to create a single predictive model.
- **In boosting, trees are grown sequentially: each tree is grown using information from previously grown trees.**

**Boosting** for regression:

1. Set $\hat{f}(\boldsymbol{x}_i) = 0$ and $r_i = y_i$, $1 \leq i \leq n$
2. For $1 \leq b \leq n_B$:
    1. Fit tree $\hat{f}_b$ with $d + 1$ nodes with $\boldsymbol{x}$ and $r$
    2. Update $\hat{f} \leftarrow \hat{f} + \lambda \hat{f}_b$ where $\lambda$ shrinks $\hat{f}_b$
    3. Update $r_i \leftarrow r_i - \lambda \hat{f}_b(\boldsymbol{x}_i)$
3. Output $\hat{f} = \lambda \sum_{b=1}^{n_B} \hat{f}_b$

$n_B$: number of boosted trees

$\lambda$: shrinkage parameter.

How much does $\hat{f}_b$ contribute to $\hat{f}$

$d$: number of nodes in the tree.

How granular is $\hat{f}_b$?

# Improvement 3: Random forests

## Bagging + decorrelating trees

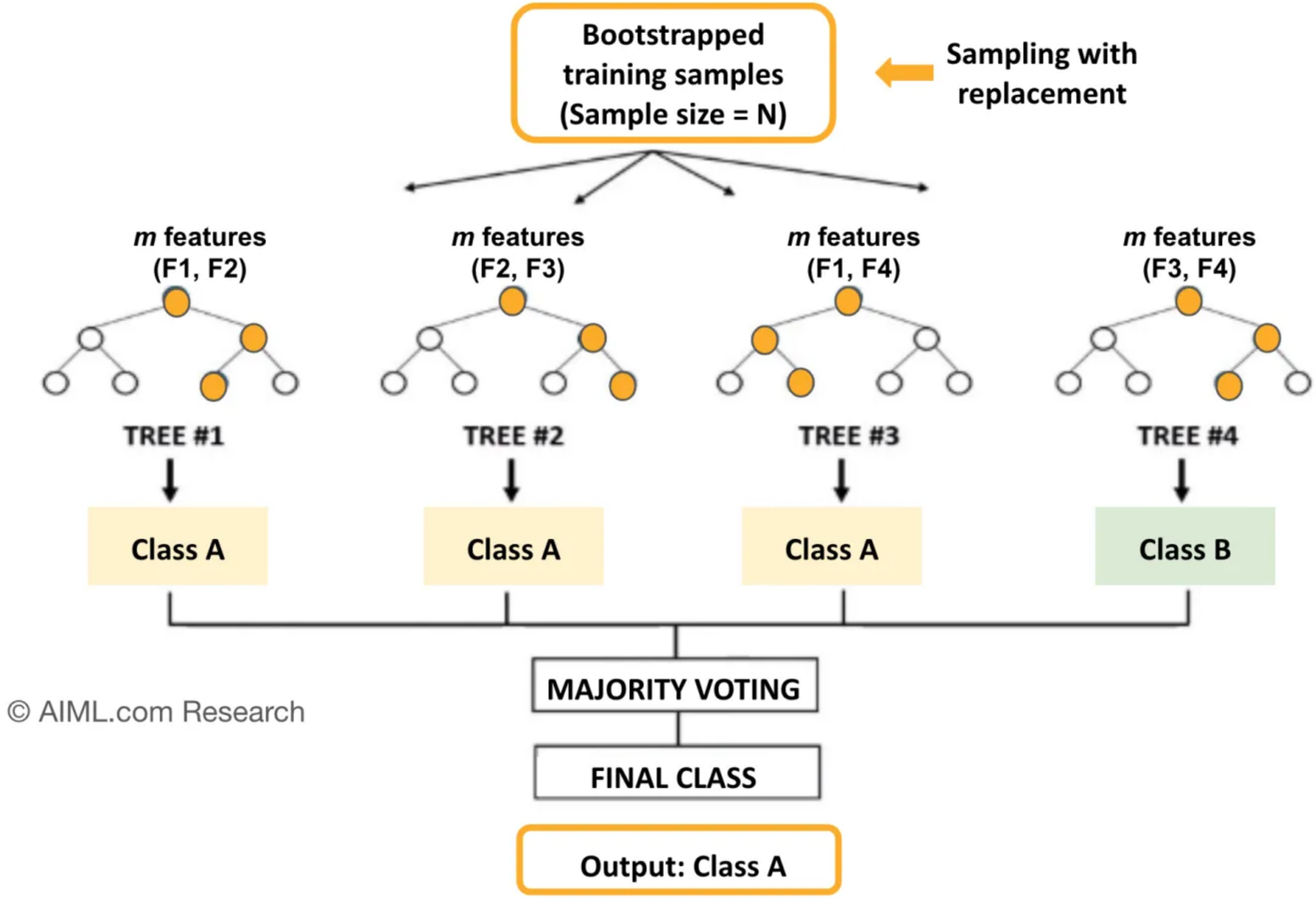
- Challenge with bagging: when predictions are correlated, then the gain is limited (if one model $\hat{f}_b$ makes a mistake, other model $\hat{f}_{b'}$ likely to make mistake too…)
- So we should probably allow for variation in models as well - we do this by allowing different subsets of covariates/features within each base model.
- When building trees, each time a split in a tree is considered, a random selection of $m$ predictors is chosen as split candidates from the full set of $p$ predictors. The split is allowed to use only one of those $m$ predictors.
- Typically $m \approx \sqrt{p}$

# Improvement 3: Random forests

## Bagging + decorrelating trees

**Training Data**
*(Sample size, N= 6, No. of features, F = 4)*

| F1 | F2 | F3 | F4 | Y |
|---|---|---|---|---|
| 2.1 | 0 | 400 | -9 | A |
| 3.0 | 1 | 890 | -42 | B |
| 2.2 | 1 | 929 | 0 | B |
| 4.0 | 0 | 324 | -23 | A |
| 3.5 | 1 | 333 | -15 | A |
| 6.0 | 0 | 215 | -9 | A |

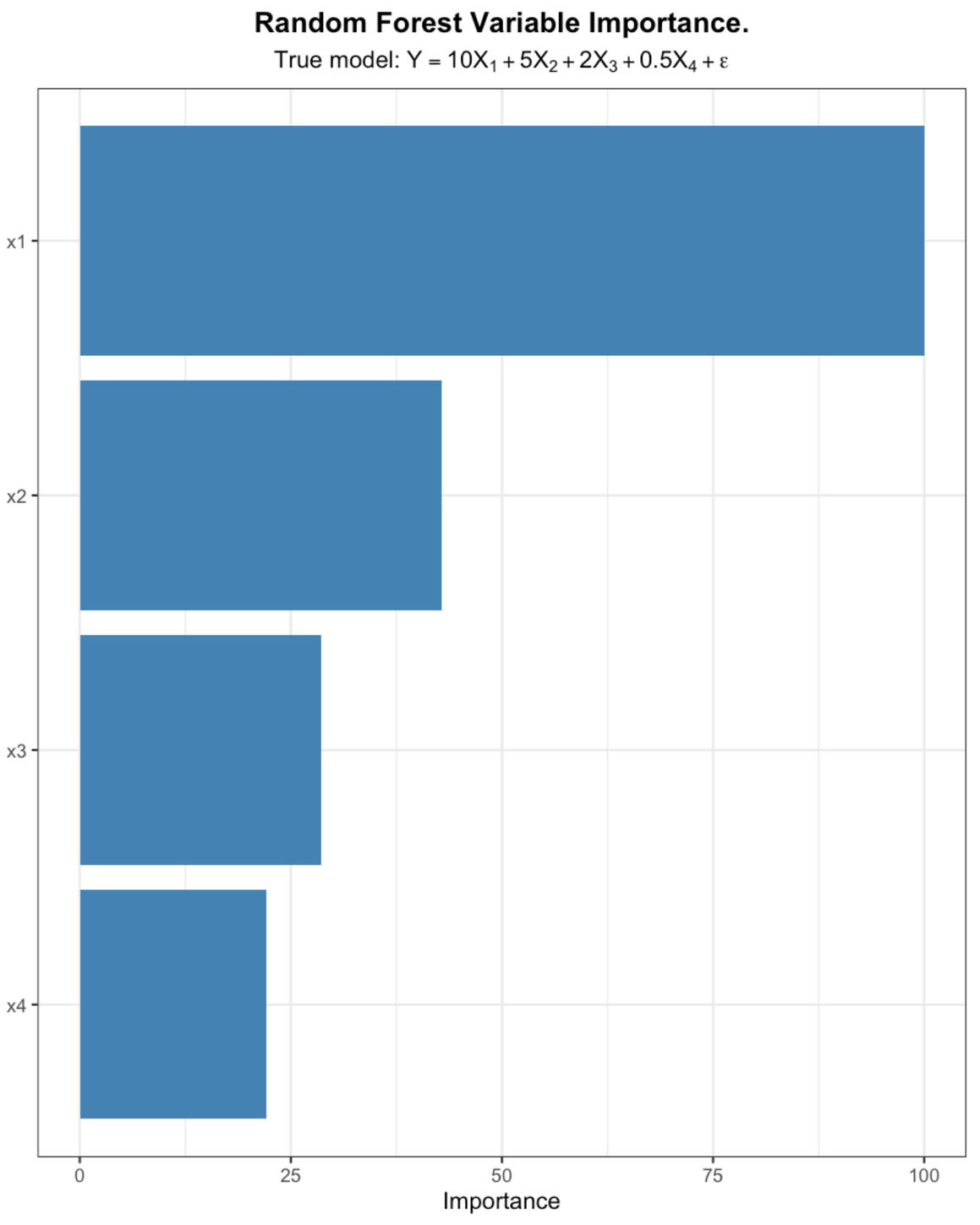



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# Variable importance

- For bagged/RF regression trees, we record the total amount that the $RSS$ is decreased due to splits over a given predictor, averaged over all $B$ trees. A large value indicates an important predictor.
- For bagged/RF classification trees, we add up the total amount that the Gini index/ entropy is decreased by splits over a given predictor, averaged over all $B$ trees

Can do variable importance for boosted trees as well using permutation importance.



# Post-class debrief

## Today we discussed…

- Decision trees are simple and interpretable models for regression and classification
- However they are often not competitive with other methods in terms of prediction accuracy

- Bagging, random forests and boosting are good methods for improving the prediction accuracy of trees.
- They work by growing many trees on the training data and then combining the predictions of the resulting ensemble of trees.

- The latter two methods— random forests and boosting— are among the state-of-the-art methods for supervised learning.
- However their results can be difficult to interpret.


That's all Folks!